

Sabrina Valerjev

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Available for full-time roles starting June 2027

Education

Northeastern University, Khoury College of Computer Sciences | Boston, MA Sept 2023 – Present
Candidate for a B.S. in Computer Science, concentration in Software Engineering Expected May 2027

Honors: Dean's List | Dean's List Scholarship **GPA:** 3.74/4.0

Coursework: Software Engineering | Network Fundamentals | Programming in C++ | Computer Systems | Programming Languages | Object-Oriented Programming | Data Science | Cybersecurity | Databases | UX Design

Activities: Aerospace Club Project Lead | Robotics Club | ColorStack

Technical Knowledge

Languages: Python | Java | C/C++ | SQL | TypeScript | JavaScript | R | Kotlin | Racket | Assembly | Ruby
Tools: Docker | Kubernetes/MiniKube | Airflow | MinIO | PostgreSQL | Redis | Honeycomb | ECS | SolidWorks
Libraries: PyTorch | Jupyter | React | Ultralytics/YOLO | OpenCV | NumPy | Pandas | Pydantic | Pyp2p | Kivy
Hardware: Arduino | Raspberry Pi | Soldering | Laser Cutting | Electrical Wiring | Circuit Design
Clearance: Top Secret Security Clearance

Experience

Software Engineering Co-op | **Tines** | Boston, MA July 2026 – Present
- Investigate and resolve backend bugs across a Ruby/TypeScript codebase as part of Team Titanium, engaging with product teams to communicate fixes, trade-offs, and workarounds
- Designed a PostgreSQL cursor-based tracking system for cleanup and action-log monitoring jobs, eliminating repeated, skipped, and stuck processing and reducing alert timeouts and misfires; presented the improvement to the engineering team
- Building a new ECS service to migrate select jobs off a shared service and prevent queue starvation, using Honeycomb and Redis for monitoring and reliability

Full-Stack Software Engineering Co-op | **MORSE Corp.** | Cambridge, MA July 2025 – Dec 2025
- Collaborated with full-time engineers on an AI-based project to enhance defense systems
- Built frontend to backend features for internal applications, and used ML tooling for model testing and evaluation
- Created a new AI model for testing, reworked our test and evaluation pipeline, and began implementing a new structure dedicated to training test data

Software Engineering Intern | **CACI International** | Goleta, CA June 2022 – Aug 2022
- Implemented Python drivers for a WindFreak Microwave Generator and a Signal Hound SM200C spectrum analyzer
- Tested driver code with a Keysight Signal Analyzer and Spike GUI
- Presented final work to the local office — CACI distributed the code to customers for applicable use of the hardware

Projects

Project ATLAS – Aerospace Club | *Python, Satellite APIs, Pydantic* | [ATLAS](#) Aug 2026 – Present
- Co-leading a ~6-person team building an open-source tool to visualize decades of global satellite imagery across 13 satellites, enabling natural-language queries such as tracking Amazon deforestation or Arctic glacier retreat over time
- Developing the data acquisition and cleaning pipeline to normalize imagery from multiple satellite sources into a consistent format for downstream modeling
- Planning an AI/ML model for change-detection analysis and a React frontend for public search and visualization

Project LEGOLAS – Aerospace Club | *Python, OpenCV, Roboflow* | [LEGOLAS](#) Jan 2025 – May 2026
- Co-led a small team to train an AI model and build a tracking camera that follows rockets during launch for more consistent flight visuals
- Delivered a well-trained model and tracking camera used across multiple semesters of club launches, extending tracking speed and capability before wrapping the project

Co-op Connect Project | *Python, SQL* | [Co-op Connect](#) Sept 2024 – Dec 2024
- Collaborated with a team in Databases to reimagine the NUWorks website to better serve its users
- Built a REST API with Streamlit, containerized with Docker, and generated mock data for testing, presenting results to class and faculty

Kinetic Maze Game | *Python, Raspberry Pi, Electrical Wiring, CAD* Aug 2022 – May 2023
- Rebuilt an unreliable prototype into a fully functional gesture-controlled game on Raspberry Pi with custom electrical wiring and P2P communication between devices; presented live at SXSW EDU (Austin, TX, 2023)

More on this and other Dos Pueblos Engineering Academy (DPEA) Projects portfolio: [DPEA-Projects](#)